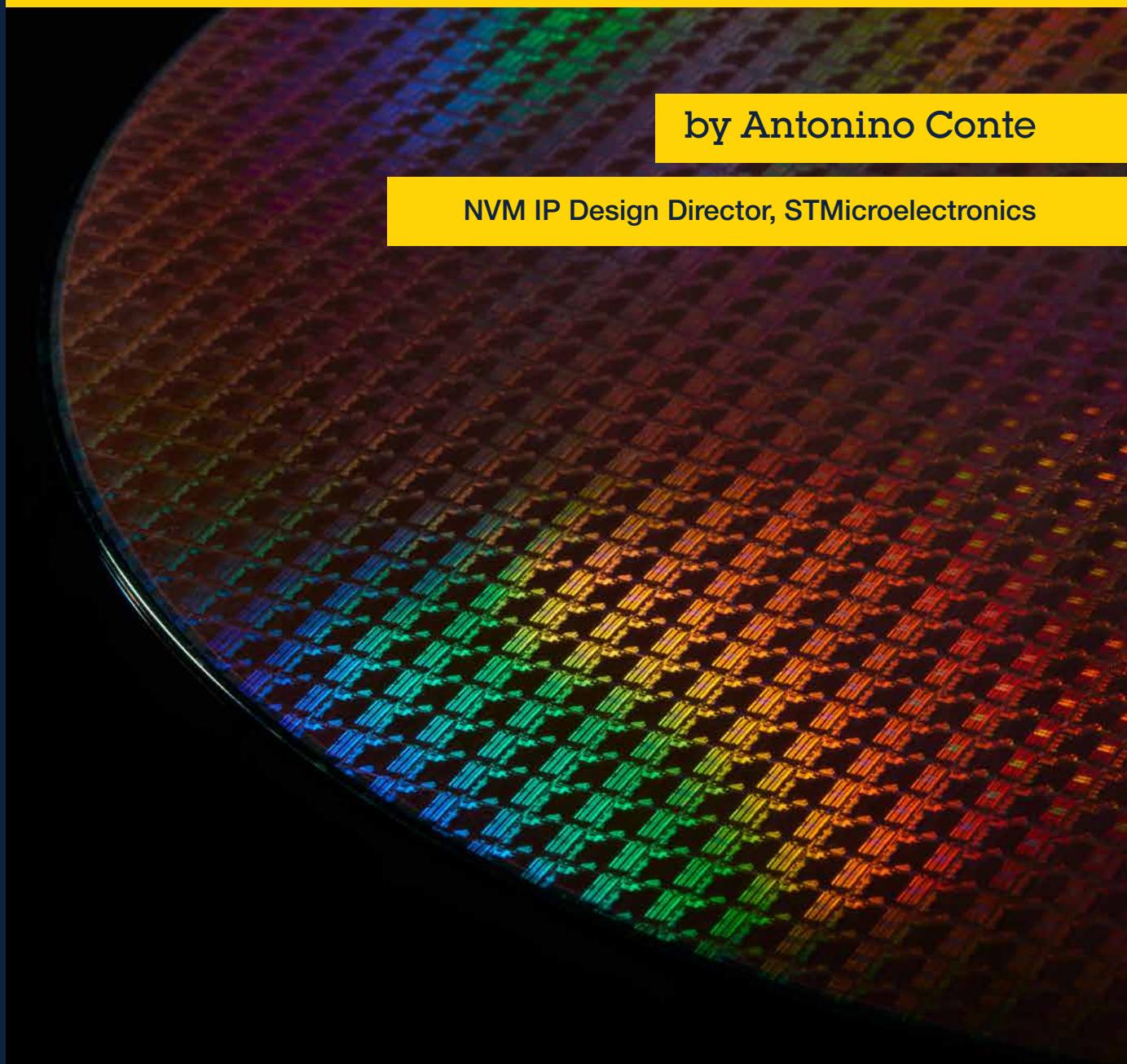




Maximizing MCU performance with 18 nm FD-SOI and embedded phase-change memory

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INTRODUCTION

Advanced CMOS technology with embedded non-volatile memory (NVM) is a key enabler for successful microcontroller system-on-chip (SoC) developments across crucial market segments, such as general-purpose, secure, and automotive applications. Several complex analog and digital functions must be integrated into the same device, each bringing its own constraints. Therefore, various factors must be considered when selecting the best technology platform.

Microcontroller SoC design requires an optimal balance of diverse requirements. These include advanced CMOS technology for high-density digital integration and low static leakage to maximize battery life. Additionally, it demands versatile device options for efficient analog

implementation, high speed with low dynamic power consumption, and a highly efficient, feature-rich NVM solution characterized by high density, compact area, low cost, fast operation, low power, and robust reliability. A CMOS platform embedding back-end NVM can meet all the requirements for efficient microcontroller design only if the NVM solution integration occurs with add-on modules that strike the best compromise in terms of complexity and cost.

This paper examines how the 18 nm fully depleted silicon-on-insulator (FD-SOI) technology with **embedded phase-change memory (ePCM)** and a bipolar junction transistor (BJT) selector provides an optimal trade-off among these requirements (see Figure 1) to meet the needs of next-generation microcontrollers.

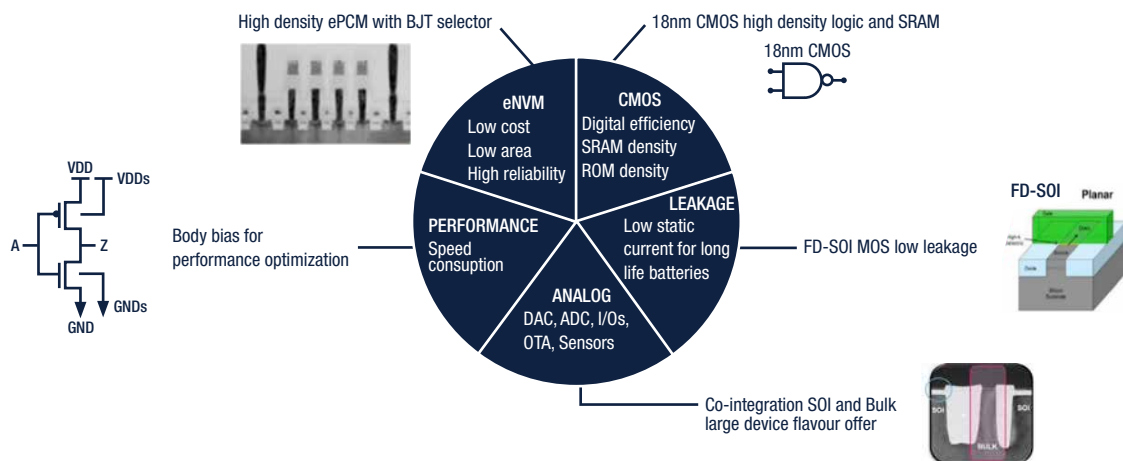


Figure 1: Microcontroller technology platform in 18nm FD-SOI with ePCM

KEY FEATURES AND BENEFITS

- **High memory density:** Thanks to the use of a BJT selector, ePCM packs more memory cells into a smaller area, allowing for larger memory capacity without increasing chip size.
- **Cost-effective integration:** ePCM adds non-volatile memory to FD-SOI CMOS chips without significantly increasing manufacturing complexity or cost, making it an economical choice.
- **Reliable and durable:** It offers excellent data retention for up to 10 years and can handle up to 100,000 write cycles, ensuring long-lasting and dependable memory performance.
- **Fast and efficient reading:** ePCM reads data very quickly (under 10 nanoseconds) while consuming very little power during read operations, which helps save energy.
- **Good writing speed:** It supports a competitive write speed (~3 megabits per second), balancing performance and power consumption.
- **Simplified software control:** Because bits can be changed in both set and reset states, managing memory through software is easier and more flexible.

18 nm FD-SOI TECHNOLOGY FOUNDATIONS

The driving factors of FD-SOI technology

The challenges associated with CMOS technology scaling are primarily linked to the minimum transistor channel length (L) achievable at a specific node. This parameter is essential to increase the density of digital ports per mm^2 . Higher transistor density allows for more functions to be integrated into the same device and inherently results in an increased clock frequency. However, this also leads to higher static and dynamic current consumption. As a result, advanced CMOS technologies necessitate a reduction in power supply voltage and all related energy consumption parameters, including static leakage.

Ultimately, transitioning to an advanced CMOS technology node (reducing gate length L), the physical properties of CMOS devices must comply with various requirements:

1. Reduction of maximum power supply (V_{DD}) and threshold voltage (V_{TH}).
2. Reduction of gate oxide thickness to improve CMOS speed.
3. Containment of transistor leakage, including gate leakage, drain-induced barrier lowering (DIBL), and gate-induced drain leakage (GIDL).

In recent years, these requirements have paved the way for significant process changes aimed at improving the electrostatic control of MOS transistors with a minimum length less than 32 nm.

Starting from 28 nm, the scaling limits of bulk technologies, such as 90, 65, and 40 nm which are widely used in microcontrollers, have become increasingly apparent. Consequently, the adoption of fully depleted devices has become essential to overcome these limitations. In this context, two major trends have emerged: FinFET and FD-SOI (see Figure 2).

In both cases, the key feature is the introduction of a thin silicon body that is fully depleted, with source and drain terminals isolated by a silicon oxide layer. This isolation prevents any parasitic current path, thereby improving the electrostatic control of the channel, despite a significant difference in implementation (3D in FinFET versus 2D in FD-SOI). Figure 3 illustrates how the 3D FinFET can be conceptually transformed into an equivalent 2D FD-SOI.

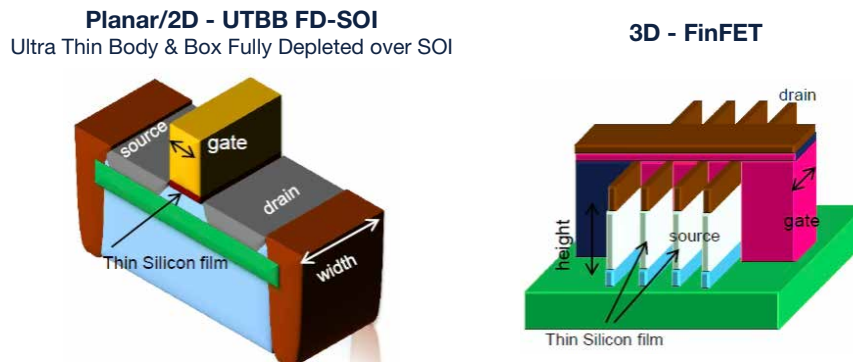


Figure 2: Fully depleted devices in FD-SOI planar and FinFET 3D technologies

The differences between the two solutions are primarily based on costs and density capabilities. The FD-SOI process offers better compromise in terms of process complexity at moderate performance levels, while FinFET enables very high-density CMOS and high performance at a higher wafer cost.

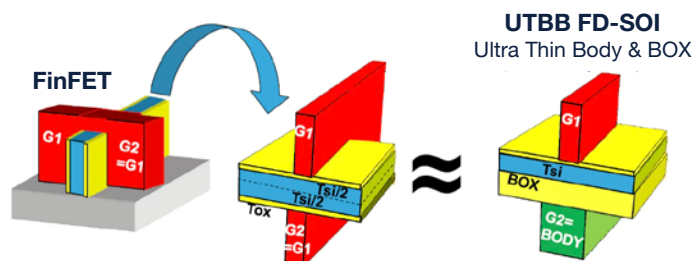


Figure 3: FinFET / FD-SOI equivalence

HK metal gate introduction and implications

As the CMOS minimum gate length dimension decreases, the gate oxide must also be reduced to recover electrostatic control of the MOS channel through the vertical electric field. This reduction, up to the 40 nm node, has been achieved using the classical SiON/poly gate scheme (SiON = silicon oxynitride), which has been adequate for gate oxide thicknesses in the range of approximately 24 Å.

Below the 40 nm node, the need to further reduce the dielectric thickness conflicts with an increasing gate leakage current, which becomes significant for microcontroller applications.

To address this issue, the SiON dielectric has been replaced by a high-k dielectric (Hf-based dielectric), which has a higher relative electrical permittivity (ϵ_r). This replacement allows for the reduction of the equivalent oxide thickness (EOT) while significantly reducing the gate leakage current by two orders of magnitude, as shown in Figure 4.

The adoption of high-k dielectric further increases the complexity of gate composition, as new challenges at the interface between the high-k dielectric and the poly-silicon gate must be addressed. One major issue with this new scheme is the need to resolve the so-called "Fermi level pinning," which prevents the control of the transistor threshold (NMOS and PMOS) through the proper selection of the gate's work function via its material composition, as is typically done.

WHAT IS FERMİ LEVEL PINNING AND WHY IS IT IMPORTANT?

Fermi level pinning limits the ability to control the transistor's threshold voltage by fixing the energy level at the metal-semiconductor interface. This restriction reduces design flexibility, making it difficult to optimize transistor performance, especially in advanced, scaled CMOS technologies.

Overcoming Fermi level pinning is essential to achieve low leakage, high speed, and precise control of transistor behavior, which is why adopting high-k metal gate technology is crucial for modern semiconductor nodes like 18nm FD-SOI.

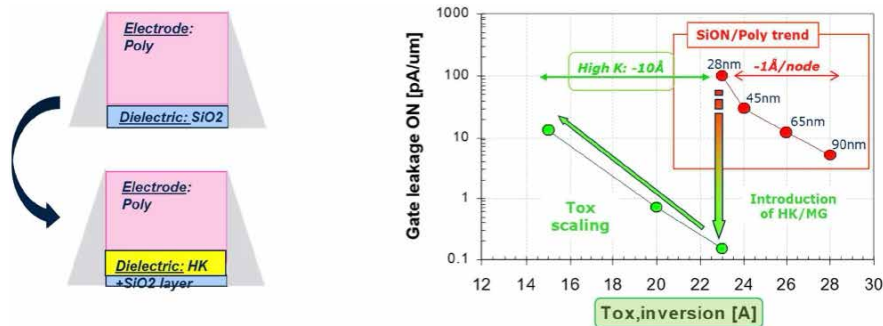


Figure 4: Reduction of gate leakage current with high-K dielectric

The "Fermi level pinning" phenomenon has been extensively modeled and studied, leading to the conclusion that adopting a metal gate in the gate stack of high-k CMOS technologies is a viable solution. To conclude, the scaling of CMOS technology requires the adoption of high-k dielectric with the addition of metal gate, bringing to the so called "high-k metal gate solution", which solve both gate leakage and Fermi level pinning. The final composition of the metal gate stack is illustrated in Figure 5.

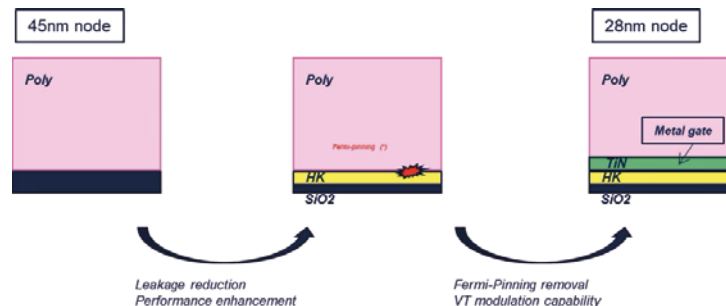


Figure 5: From SiON to high-K metal gate

FD-SOI transistor performance

One of the major advantages of FD-SOI, as previously noted, is the improved electrostatic control of the transistor channel and the mitigation of parasitic effects otherwise present in bulk technology.

The most important structural improvements of the FD-SOI MOS include:

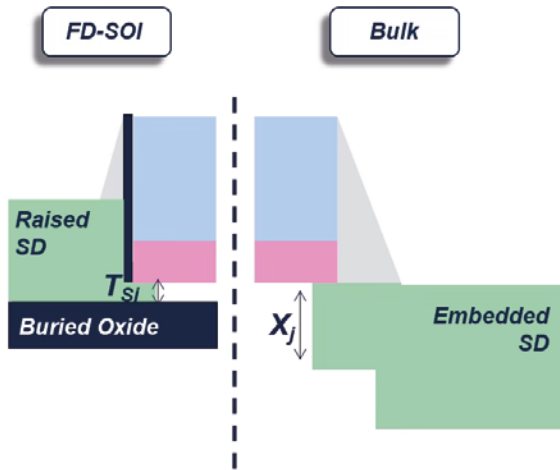
- **Gate leakage:** reduced by a factor of 100, thanks to the high-k metal gate.
- **DIBL (drain-induced barrier lowering):** reduced by about 50mV for the same node.
- **GIDL (gate-induced drain leakage):** mitigated by the removal of the pocket implants (thin film property).
- **SCE (short-channel effect):** mitigated thanks to the fully depleted channel.

The detailed description of above benefits is beyond the scope of the present paper; however, it is worth mentioning how DIBL effect is improved thanks to the MOS architecture in FD-SOI.

Figure 6 compares the structural differences between the FD-SOI (on the left) and the Bulk MOS (on the right). It can be noticed through the reported formulas that the reduction of the DIBL effect in FD-SOI is controlled by the silicon thin film thickness (T_{Si}), whereas in bulk technologies, it is controlled by the junction depth (X_j), which is significantly larger.

Other technological features further enhance the intrinsic advantages of FD-SOI:

- **VTH MOS modulation:** achieved through the FBB terminal.
- **Boosting of MOS performance:** thanks to strain effects.
- **Analog transistor matching:** improved versus bulk.
- **Latch-up and SEL (single event latch-up) robustness:** intrinsic property of FD-SOI.



$$\begin{aligned} \text{BULK} \quad DIBL &= 0.8 \frac{\epsilon_{Si}}{\epsilon_{ox}} \left(1 + \frac{X_j^2}{L_{eff}^2} \right) \frac{T_{ox}}{L_{eff}} \frac{T_{dep}}{L_{eff}} V_{ds} \\ \text{FD-SOI} \quad DIBL &= 0.8 \frac{\epsilon_{Si}}{\epsilon_{ox}} \left(1 + \frac{T_{Si}^2}{L_{eff}^2} \right) \frac{T_{ox}}{L_{eff}} \frac{T_{Si}}{L_{eff}} V_{ds} \end{aligned}$$

Figure 6: Comparison of DIBL effects in bulk CMOS versus FD-SOI technologies

FBB V_{TH} MOS modulation

Figure 7 illustrates the complete FD-SOI MOS cross-section with bulk terminal access. In FD-SOI technology, the bulk terminal enables precise control of the MOS transistor threshold voltage (V_{TH}) over a wide range, through both forward body bias (FBB) and reverse body bias (RBB). This distinctive capability allows optimization of the trade-off between speed and leakage for any process variation—from slow to fast digital corners—by dynamically adjusting the transistor's threshold voltage.

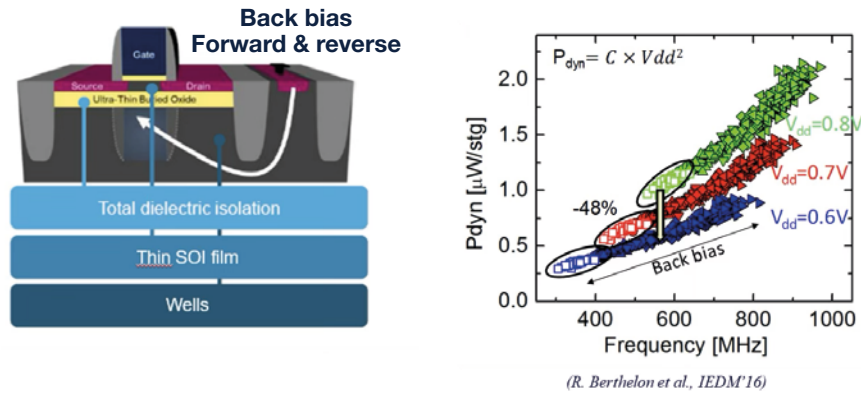


Figure 7: Beneficial effects of Forward Body Bias (FBB) and Reverse Body Bias (RBB) control in FD-SOI technology

Process dispersions can be compensated with the appropriate combination of VDD (low-voltage power supply) and FBB to improve performance on slow devices and reduce leakage on fast ones.

Strain boosting effect on MOS gain performance

The MOS transistor gain performance can be modulated by changing the effective mass of the hole and electron particles, and consequently their mobility, which is directly linked to the gain of the transistors. This effect can be achieved by inducing either compressive or tensile stress in the transistor channel. In 18 nm FD-SOI with ePCM technology, the adoption of a different orientation of the MOS device channel ($\langle 110 \rangle$ instead of $\langle 100 \rangle$, used up to 28 nm node) and the use of SiGe in the PMOS channel allowed the generation of compressive strain in PMOS and tensile strain in NMOS low-voltage transistors, thus boosting majority carrier mobilities, as reported in Figure 8.



Figure 8: Strong improvement of low-voltage PMOS induced by SiGe compressive strain

FD-SOI analog matching

It is worth mentioning the matching properties of 18 nm FD-SOI CMOS for analog design applications. In bulk technologies, analog matching is primarily determined by random process variability, typically identified in three different types: poly grain granularity (PGG), random dopant distribution (RDD), and line edge roughness (LER). These factors all impact the MOS threshold and gain. In Figure 9, a representation of these components and their impact on the surface potential distribution (Φ_s) under the gate is shown. This distribution is directly involved in determining the MOS transistor threshold.

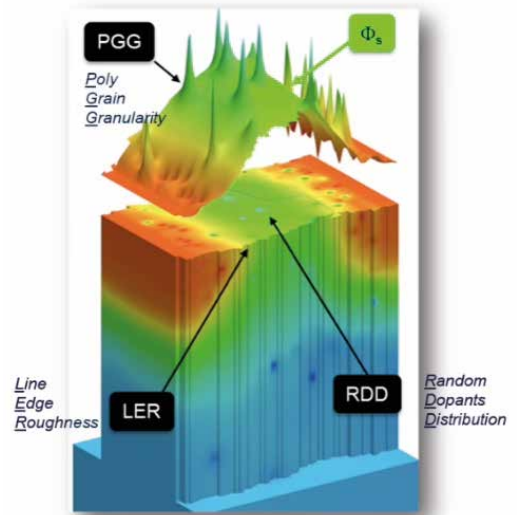


Figure 9: Atomistic simulations highlighting variability sources and impact on Φ_s

In this process, PGG and RDD are absent by design, while LER has a reduced impact. This absence benefits the analog matching properties of CMOS transistors. However, a new source of mismatch, not present in bulk processes, must be considered. This mismatch is correlated with the control of the thin silicon film thickness variability, which depends on wafer fabrication quality. Local variations in silicon film thickness induce local variations in the MOS transistor threshold voltage (V_{TH}). This new effect, named TSI, must be addressed in analog design using proper layout techniques.

Latch Up and SEL robustness

In the intrinsic MOS structure of FD-SOI, source and drain terminals are isolated from the bulk by the presence of the ultra-thin buried oxide (as indicated in Figure 7). Thanks to this unique characteristic of FD-SOI technology, phenomenon like latch up, which is based on the activation of a parasitic BJT positive feedback loop, is suppressed. For the same reason 18 nm FD-SOI with ePCM technology is extremely robust versus SEL phenomenon (single event latch-up).

Hybrid integration of bulk devices

This technology completes its broad range of device solutions by offering SOI and bulk co-integration, thus providing large possibility of analog design optimization (Figure 8). MOS transistors for 3.3 V range operability and other bulk devices (varactors, diodes, resistors ... etc.) are available for implementation with SOI devices in the same circuitry.

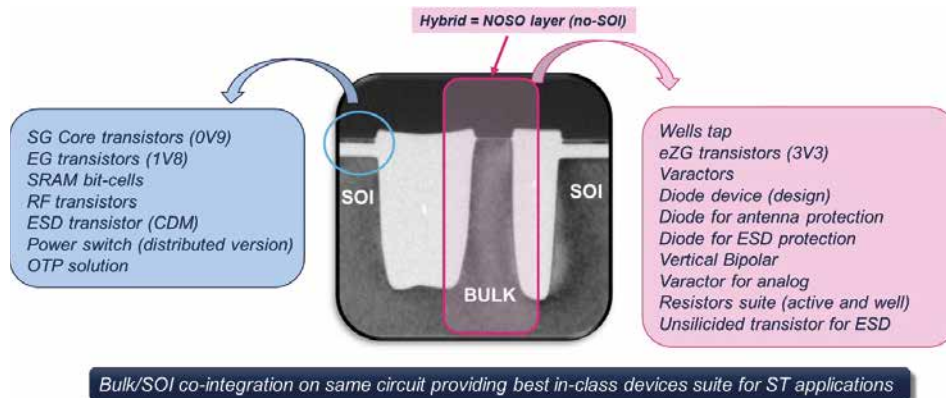


Figure 10: Hybrid co-integration of bulk and SOI technologies

EMBEDDED PHASE-CHANGE MEMORY TECHNOLOGY

The driving factors of phase-change memory

PCM belongs to the new category of resistive memories, which are significantly differentiated from floating-gate memories in terms of the storage element working principle and their impact on CMOS process integration. While any floating-gate memory (EEPROM, FLASH, NAND, ESF3, ESTM, etc.) is based on the change in the sensing transistor threshold induced by a charge injected or removed from a floating gate, acting as a series capacitor, all resistive memories (PCM, ReRAM, MRAM, among the most important) are based on the information status "high resistance vs. low resistance," which is stored in an element built in the back-end. Here, the back-end refers to the part of the silicon process above the contact layer, including metals.

This is a critical aspect of all resistive memories, which is an advantage over floating-gate memories, especially considering the added cost in the overall process induced by the introduction of the "memory module."

Being "added" in the back-end, all resistive memories, at least in principle, do not have a major impact on the way the CMOS front end is constructed. This is particularly relevant given the already higher level of complexity induced by the need for high-K metal gates in all advanced CMOS, which are mandatory to achieve the expected performance. On the contrary, all floating-gate memories require a dedicated module added to the CMOS in the front-end part of the process flow. This module must be completely decoupled from the high-K metal gate, thereby adding cost in terms of process steps and the number of required mask layers.

In other words, the migration to resistive memories in advanced embedded CMOS has been perceived as mandatory starting from the 28 nm node and below. The 28 nm node is considered a turning point in this regard, with some key silicon providers still working on bulk technologies and others migrating to FD-SOI or FinFET.

Phase-change memory working principle

The working principle of PCM relies on the unique capability of GST material, an alloy with a stoichiometric ratio of germanium, antimony, and tellurium, to change its resistivity through the Joule effect modulated by the injection of a current pulse with the appropriate shape and intensity.

In Figure 10, the basic structure and working principle of the PCM storage element are schematically represented. The fundamental structure of the PCM storage element consists of a bottom electrode connected to a heater, surmounted by an unconfined phase-change layer (constructed with GST alloy), and followed by a top electrode.

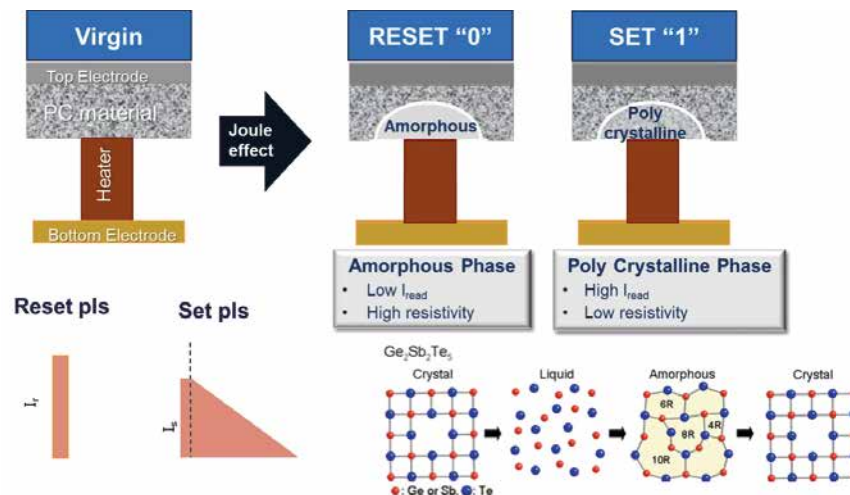


Figure 11: Phase-change storage element structure and working principle

The two written states of the GST are amorphous (high resistance or reset) and crystalline (low resistance or set), with crystalline representing the stable condition that the material tends to achieve. The GST can change status from amorphous to crystalline and vice versa through the application of a current pulse of appropriate height and shape, but always in the same direction (or polarity). This is a key point differentiating the PCM cell from other resistive NVMs and providing a competitive advantage, as explained below.

The amorphous state is achieved by applying a rectangular current pulse (reset pulse) for a short duration (in the range of 100 ns) and with a current value typically higher than the pulse used to transition to the crystalline state. During the application of the reset pulse, the GST reaches its melting point and is then rapidly cooled due to the fast current drop, bringing its atomic composition into a disordered state with a significant number of defects in the energy band diagram. These defects are essential for the conduction mechanism in this state.

The crystalline state, on the other hand, is obtained by applying a trapezoidal current pulse (set pulse) with a current maximum value below that of the reset pulse and a negative ramp with a well-controlled slow slope, which is essential to achieve an ordered atomic composition.

The virgin state of the GST out of the fabrication process is crystalline, but the material composition is not suitable for cell operation. During the fabrication process, the material contains aggregates or grains of germanium, which must be locally expelled from the zone physically close to the heater to create a “dome” with the correct composition (as shown in Figure 10).

To transition from the virgin state to the operative condition, it is mandatory to prepare or “activate” the material and the dome of each cell by applying a “forming pulse.” This is a trapezoidal current pulse with a current height typically of the same order as the reset pulse. The result of this operation is the expulsion of excess germanium from the dome and the creation of the desired local composition. The forming operation is performed once in the device's lifetime during EWS (electrical wafer sorting) in a controlled testing environment to minimize testing time.

Retention and reliability demand

Like other non-volatile memory (NVM) technologies, embedded phase-change memory (ePCM) must meet the highest quality standards under aggressive environmental conditions and severe cycling and retention requirements, as dictated by various application profiles such as automotive and industrial, which are among the most demanding.

Typical industrial reliability requirements, for example, are differentiated between “Code” and “Data”:

- **Code:** high memory capacity (from 1 to 32 MB in automotive), low to moderate cycling requirements ranging from 1 Kwrite and 10 years of retention after cycling at 140 °C. Example: program application code.
- **Data:** low memory capacity (128 to 256 KB), high cycling demand up to 100 Kwrite, and 1 year of retention after cycling at 140 °C. Example: data logging storage.

To achieve such high-quality standards, extensive experimental and statistical data are collected on technology test vehicles. This leads to the implementation of complex writing algorithms that prevent permanent hardware damage (e.g., cell open bits or circuitry failure) and soft failures (e.g., loss of information).

Permanent damage cannot be recovered if the memory circuitry is damaged due to excessive high-voltage conditions beyond the technology's tolerance. Prevention in such cases relies on design simulations and automatic checks of the safe operating area (SOA). In cases of open bit occurrences, error correction code (ECC) techniques are widely used to ensure data integrity, even in the presence of double errors per word.

Soft failures, on the other hand, result from the loss of information in cells, either in the reset or set states. The reset state (amorphous state) is unstable for GST material, which naturally tends to crystallize over time (tens of years). This means that, after some time, bits in the reset state will transition from high resistance to low resistance. Conversely, the set state, while intrinsically more stable, is affected by a phenomenon called set drift, characterized by an increase in resistance, transitioning from low resistance to high resistance.

In the plot in Figure 12, two distributions of ePCM reset and set cell populations collected from a certain number of devices are shown (DD=Defect Density in the plot). After 100,000 write cycles and retention baking at 90 °C for up to 1,000 hours, a clear separation between the two distributions is observed, ensuring robust read operations even after extensive cycling and retention stress. No crystallization effects are detected on the reset side, and the set drift phenomenon remains fully controlled. Both distributions are presented in terms of current, as current is the physical property directly sensed by the memory circuitry under specific cell polarization conditions.

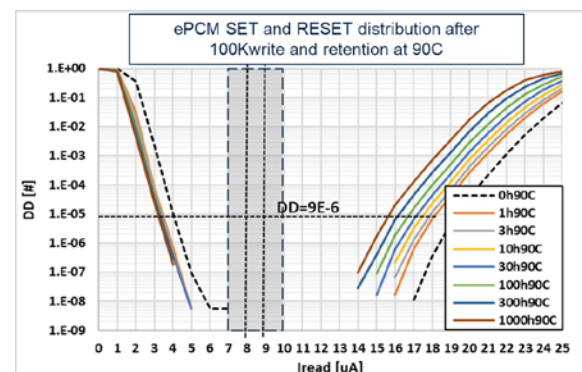


Figure 12: Example of Reset and Set distributions after 100Kwrite cycling and retention at 90°C, showing large window for reading

Writing pulses and algorithms

Three main types of current pulse sequences are managed through an algorithm embedded in the ePCM IP and fully controlled by the digital and analog circuitry present in the memory. The various sequences are distinguished between forming (previously described), power reset and set, and user reset and set. The last two are adopted in different contexts and for different purposes.

Forming operation

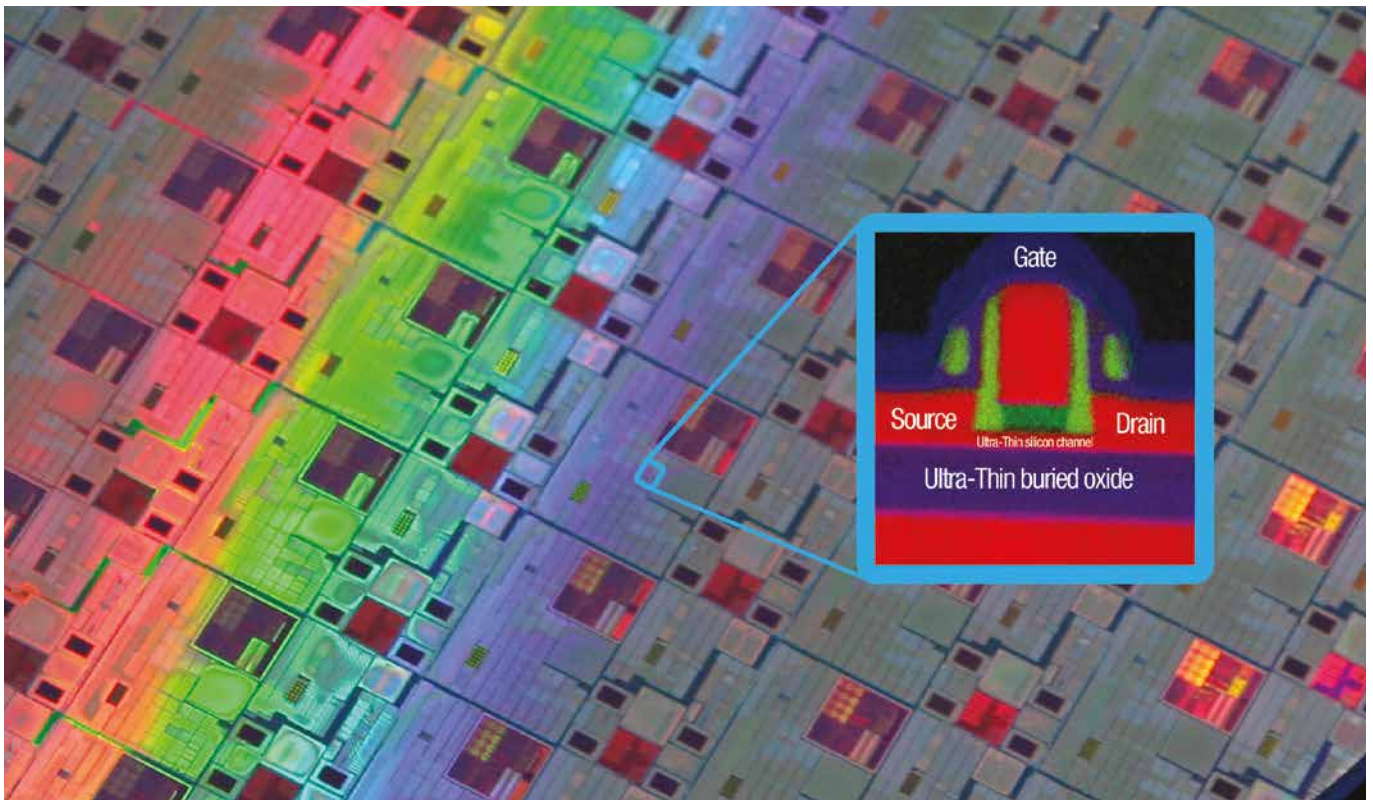
Already previously described, the forming operation is used in EWS testing to activate the material and create the dome over the heater in each ePCM cell.

Power algorithm

The power algorithm is primarily used to prevent the loss of information in PCM cells during the soldering process. Soldering conditions typically impose very high temperatures, reaching approximately 260°C for a few minutes. Such extreme temperatures may cause information loss and necessitate a higher current to write a more robust status in the cell, regardless of whether the content is Reset or Set. However, this writing sequence cannot be repeated throughout the product's lifetime, as it may result in a high rate of permanently damaged bits due to the strong current density of the Power Pulse. Nevertheless, technology allows a limited number of Power Pulse sequences to rework parts, such as firmware updates, before soldering.

WHY ARE THESE ALGORITHMS IMPORTANT?

- They **ensure data reliability and integrity** under different conditions—during manufacturing (power algorithm) and normal operation (user algorithm).
- They **prevent permanent hardware damage and soft data failures**, extending the memory's usable life.
- They enable **robust performance of the ePCM memory**, critical for demanding microcontroller applications requiring high endurance and retention.



User algorithm

The user algorithm defines the pulse sequence adopted during the product's life in the field.

The sequence consists of a certain number of reset pulses and set pulses, each with a different parameter. This approach ensures that the entire cell population is efficiently written, as the first pulse may not fully write all cells. Additional attempts are made to ensure that the cell states are well separated in terms of current.

Typically, when a word-write command is sent to the ePCM, the NVM IP performs an initial attempt by sending the first reset and set pulses to all cells requiring a status change.

After this operation, a verification step is performed by reading the word content twice against a reference current. The first Reset Verify operation compares the status of cells expected to be in the reset state against a Reset Verify current (I_{RESET}). Cells with a current higher than the reset verify current are marked as NOK. Similarly, a set verify operation compares cells expected to be in the set state against a set verify current (I_{SET}). Cells with a set current below the set verify current are also marked as NOK.

After the initial attempt, all cells marked as NOK are treated with the second reset and set pulses, and the process continues until the final pulse.

Competitive advantages of ePCM

The shift from traditional floating-gate memories (such as NOR Flash, split-gate, and ESTM) to back-end memory technologies like ePCM, ReRAM, and MRAM in advanced CMOS processes is driven largely by cost efficiency. Floating-gate memories, while proven and reliable up to the 40 nm node, add significant complexity and expense due to extra mask layers and additional process steps required in the CMOS front-end. This makes them less practical for advanced technology nodes.

The 28 nm node marks a critical transition, with most embedded memory solutions migrating towards FD-SOI or FinFET platforms, although some bulk-based solutions still exist. In this context, the embedded NVM module must be as streamlined as possible, minimizing impact on the CMOS front-end to keep costs and process complexity under control.

Bearing this in mind, in the panorama of back-end memories, ePCM is unique for various reasons:

- **Superior cell density**, in the version implemented with a BJT selector.
- **Competitive cost** in terms of the NVM added module integrated with FD-SOI CMOS.
- **High reliability performance** (10-year retention and 100K write cycles).
- **High read performance** ($t_{\text{ACC}} < 10$ ns, read dynamic consumption < 0.9 $\mu\text{A}/\text{MHz/bit}$).
- **Good write throughput** (~ 3 Mb/s).
- **Easier software management** thanks to the bit alterability both in set and reset operations.

The outstanding performance of the ePCM in terms of cell size is particularly noteworthy, as even at 18 nm, it outperforms ReRAM and MRAM, despite these two solutions being realized in more advanced technology nodes (16 nm or 12 nm).

The main reason behind this unique result is the physical property of the cell. The write operation requires reset and set current pulses of the same polarity, meaning that the current always flows in the same direction in the cell, regardless of the write operation. In contrast, other solutions require inversion of current polarity for either reset or set operations.

The immediate consequence of this property is the ability to adopt a unidirectional selector device for the ePCM cell, such as a vertical BJT. Other solutions require a MOS selector (generally a low-voltage NMOS), which provides a significant advantage in terms of cell area due to the intrinsic property of BJTs, which have higher driving capability than MOS transistors. Additionally, BJTs are much more robust devices compared to low-voltage MOS selectors during write operations. This robustness greatly simplifies the design of the memory array and the related decoding circuitry, allowing for a more compact design compared to equivalent solutions in ReRAM or MRAM implementations.

The NVM circuitry design and the ePCM cell size provide the technology with a highly competitive memory in terms of area, which is a significant advantage for all microcontroller products, particularly when increased memory capacity is required (from 1 to 32 MB and beyond). Additionally, all electrical performances are comparable to or better than other NVM solutions, thereby enhancing the competitiveness of ePCM.

Leveraging 18 nm FD-SOI with ePCM technology in MCU architecture and design

The ePCM NVM IP design differs significantly from other memory technologies due to the electrical characteristics of the BJT selector, which must be carefully considered when constructing an array of cells. A key objective in NVM design is to create an array with decoding circuitry that is as compact as possible, maximizing array efficiency. Array efficiency is defined as the ratio of the array area to the total area, including both the array and its associated circuitry:

$$\text{Array efficiency} = \frac{\text{Array area}}{\text{Array area} + \text{Circuitry area}}$$

Achieving high array efficiency is a critical goal that can ultimately determine the success or failure of the entire NVM technology.

As shown in Figure 13, selecting a word line (or page) of an array of ePCM cells induces a current injection along the word line. Each cell, when biased for either writing or reading operations, contributes to the total current.

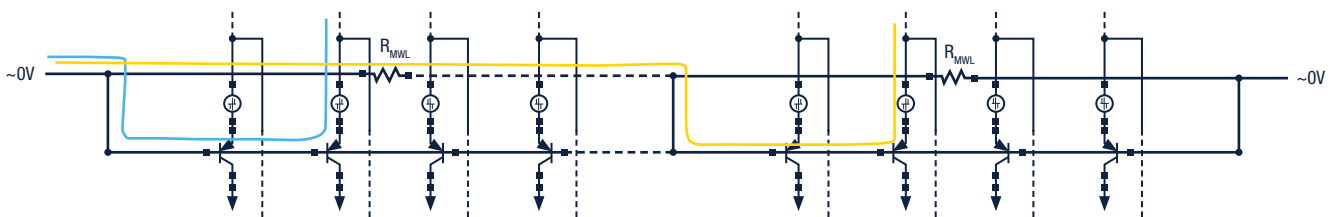


Figure 13: Assembly of ePCM cells in a word-line and current injections

Such current injections can be relevant and represent a major difference compared with other NVM array designs, such as floating gate or back-end with MOS selector, where almost no current flows along the word line. The ePCM array is assembled using segmentation techniques, distributing word-line decoding buffers instead of concentrating them in the same layout area (see Figure 14 for reference). It is clear how important the design of decoders, particularly the distributed word-line buffers, is in determining the array efficiency of the ePCM IP. Optimized techniques based on 1.8 V MOS transistors have been developed to generate a very compact decoding structure, ensuring high array efficiency in the state-of-the-art ePCM design.

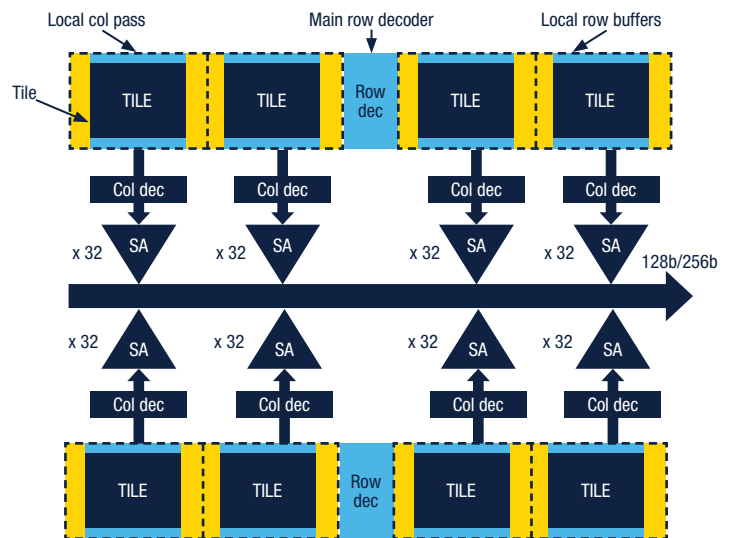


Figure 14: Array segmentation to reduce the effect of current injection along word-lines

Figure 14 illustrates the composition of the array along with the reading circuitry (sense amplifiers, SA). The circuitry is distributed and organized by tile, where each tile represents an elementary array of cells.

The 18 nm FD-SOI technology with ePCM and a BJT selector is a subsystem that includes dedicated analog and digital circuitry required for executing memory write and read operations.

The DMU is the part of the memory design physically closest to the ePCM array and includes decoders (word-line and column) to address words of cells, sense amplifiers for reading operations, and program drivers for writing operations. The SMU, on the other hand, includes all the analog blocks used to generate regulated voltages and currents for writing and reading operations. Finally, the digital controller interprets the commands sent by the SoC and sequences all writing and reading operations coded in specific ePCM algorithms.

Typical 18 nm FD-SOI with ePCM technology performance and mission profiles

This technology offers superior advantages for future microcontroller developments versus other technology platforms.

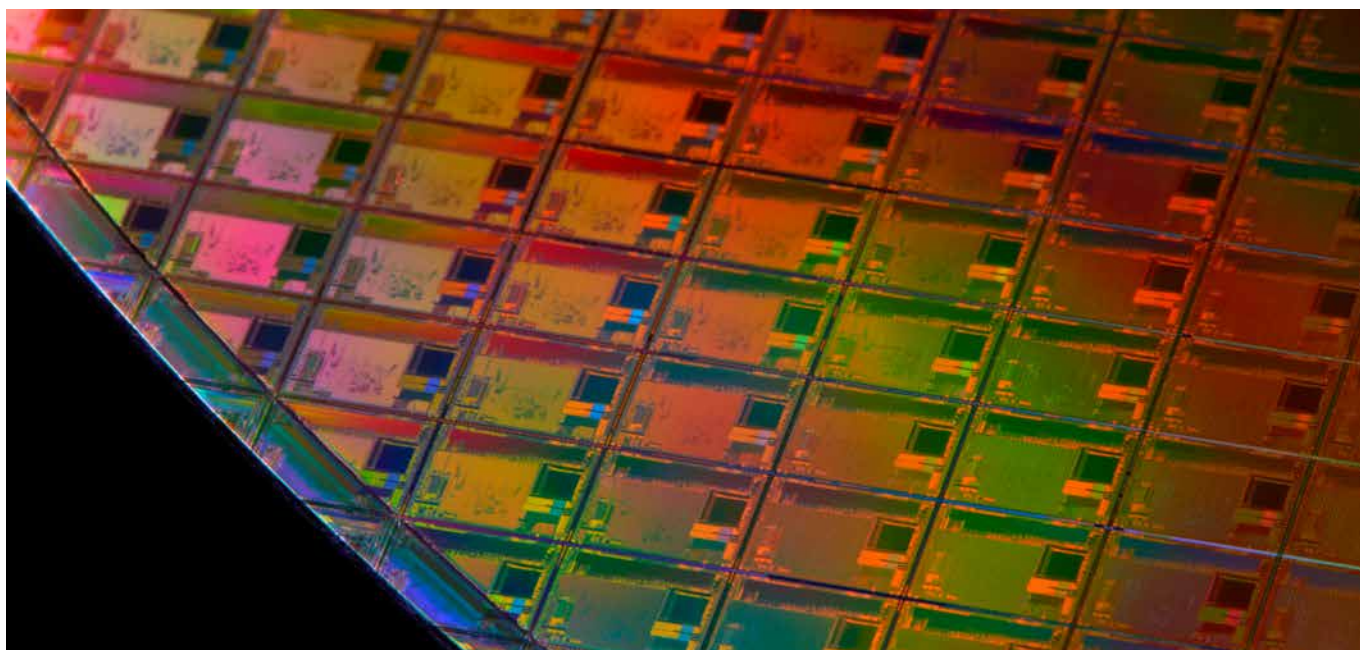
Typical performance and supported mission profiles are provided respectively in Table 1 and Table 2, where the main characteristics of the first product designed using this technology for the general-purpose market. Note that the aggressive profile fully satisfies the stringent requirements of industrial applications.

Features	Performance	Comments
Reading mode 1	128b	
Reading mode 2	256b	Reading on 2 banks (128b each)
ECC scheme	17b (2EC + 3ED)	
Word write [us]	124	50% of bits
Page write [us]	750	50% of bits
Access Time/Range [ns]	10.5ns	Worst case
Reading consumption [uA/MHz]	220uA/MHz (typ)	
Write consumption [mA]	ICC Idle + 6mA	
Temperature T _j [C]	-40/+140	Junction temperature

Table 1: Typical performance of an ePCM IP in this technology

Typical microcontroller mission profile	
Condition	Specification
External operating voltage (V _{cc})	1.62 to 3.6V
Operating temperature (T _j)	-40 to +140°C
JEDEC soldering compliancy	yes
Code memory reliability	Cycling: 1 Kwrites
	Retention: 10 years at 140°C
Data memory reliability	Cycling: 100 Kwrites
	Retention: One year at 140°C

Table 2: Mission profile for industrial perimeter



CONCLUSION

This technology combines advanced 18 nm FD-SOI CMOS with embedded ePCM non-volatile memory featuring a robust BJT selector. This fusion delivers exceptional performance and cost-efficiency, alongside the most dense and scalable embedded NVM solution available today. Together, these strengths make this technology platform ideal for the next generation of microcontrollers across all market segments.

- The FD-SOI CMOS architecture **effectively addresses leakage specifications**, while 18 nm lithography enables the integration of advanced digital standard cells and efficient SRAM, supporting complex digital functions.
- The co-integration of Silicon-On-Insulator (SOI) and bulk devices offers **a broad spectrum of analog components for optimized design**. Additionally, body bias management inherent to the FD-SOI MOS structure provides a flexible way to balance high speed with low dynamic power consumption.
- The embedded PCM with BJT selector process module **enables the integration of highly dense non-volatile memory within SoC designs**. This memory delivers excellent reliability in terms of cycling endurance and data retention, meeting stringent microcontroller requirements, along with superior speed and low dynamic power consumption achieved through advanced design techniques.

ST's 18 nm FD-SOI with ePCM technology delivers unmatched efficiency, reliability, and scalability for next-generation microcontrollers across all markets.

ACRONYMS

Abbr.	Full Form	Explanation / Context
BJT	Bipolar junction transistor	Semiconductor device used as a selector in ePCM
CMOS	Complementary metal-oxide-semiconductor	Standard semiconductor technology for integrated circuits
DIBL	Drain-induced barrier lowering	Short-channel effect impacting transistor performance
DMU	Data memory unit	Part of ePCM IP handling data storage
ECC	Error correction code	Technique to detect and correct data errors
EOT	Equivalent oxide thickness	Effective thickness of gate dielectric
ePCM	Embedded phase-change memory	Non-volatile memory technology integrated in CMOS
EWS	Electrical wafer sorting	Testing process step for silicon wafers
FBB	Forward body bias	Technique to modulate transistor threshold voltage
FD-SOI	Fully depleted silicon-on-insulator	Advanced CMOS technology with improved electrostatic control
GIDL	Gate-induced drain leakage	Leakage current in MOS transistors
GST	Germanium-antimony-tellurium	Phase-change material used in PCM cells
HKMG	High-k metal gate	Advanced gate dielectric technology reducing leakage
JEDEC	Joint electron device engineering council	Standards body for semiconductor devices
LER	Line edge roughness	Source of variability in transistor threshold voltage
MOS	Metal-oxide-semiconductor	Type of transistor used in CMOS technology
NMOS	N-channel MOS transistor	Type of MOS transistor
NVM	Non-volatile memory	Memory retaining data without power
PGG	Poly grain granularity	Source of variability in transistor threshold voltage
PMOS	P-channel MOS transistor	Type of MOS transistor
RBB	Reverse body bias	Technique to modulate transistor threshold voltage
RDD	Random dopant distribution	Source of variability in transistor threshold voltage
SCE	Short-channel effect	Degradation in transistor performance at small dimensions
SEL	Single event latch-up	Radiation-induced device failure
SMU	Service memory unit	Analog block in ePCM IP for voltage/current generation
SOA	Safe operating area	Operating limits to prevent device damage
SoC	System-on-chip	Integrated circuit combining multiple functions
SRAM	Static random-access memory	Volatile memory used for fast data storage
TSI	Thin silicon film thickness variability	New mismatch source in FD-SOI analog design
VCC	External operating voltage	Supply voltage for device operation
VDD	Power supply voltage	Voltage powering the CMOS circuits
VTH	Threshold voltage	Voltage at which a transistor switches on